



Canadian Reduced Gravity Experiment Design Challenge

CAN-RGX IX (2025 – 2026)

Application Form

Notice to Applicants: Please read the Application Guide on the website in full before completing this application form. You must answer all fields as indicated. Your application may not be considered if there is missing or incomplete information. The application **MUST** be submitted by **11:59 PM EST on Sunday, October 19, 2025**. If you have any questions, please contact canrgx@seds.ca.

SECTION 1 – TEAM INFORMATION

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Institution: University of Toronto

Primary Point of Contact

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On the following page, please complete the information for every student member on the team. **Proof of enrollment is required for each participant.** Suitable proofs of enrollment include verification of enrolment letters on university letterhead or photos of a valid Student ID card. Please attach with the completed application. At least one member of the team must provide proof of active SEDS-Canada membership if accepted for the competition.

SECTION 2 – PROJECT SUMMARY

Project title (max 10 words): Crystallization in zero-G

Project Field: Technology demonstration

e.g., life sciences, physics, technology demonstrations

Lay abstract (150 words max):

Crystals grown in microgravity offer superior samples for drug development, as the absence of gravity-driven convection and sedimentation produces larger, higher-quality crystals with fewer defects than those grown on Earth. However, conventional microgravity crystallization typically requires several days and access to orbital platforms. Laser-induced crystallization provides a faster alternative, triggering crystal formation within seconds through rapid pressure drops generated by laser-induced bubbles that initiate nucleation. In this study, we use parabolic flight experiments to investigate this process under both microgravity and hypergravity conditions. By comparing lysozyme crystals formed in each environment, we aim to demonstrate the feasibility of laser-based, high-throughput crystal formation in microgravity and to quantify how gravity affects crystal size, structure, and purity. This approach could substantially reduce the cost of producing microgravity-grown crystals by leveraging short-duration suborbital flights. The research also supports Canada's leadership in a rapidly expanding market projected to exceed USD 450 million by 2033.

Team Members. A minimum of 4 team members is required. The majority of the team should be composed of undergraduate students with no more than 1/3 of the team composed of graduate or high school students.

First and Last Name	Institution	Degree in progress	Level of study	Department	Active SEDS Member?	Role on the team	Email
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Notes:

- Role on the team examples: team lead, electrical, structural, science, outreach, advisor, etc.
- At least one team member must be identified as “Team Lead”

Eligibility Requirements

We've reviewed and understood the experimental environment described in this section. In addition, we confirm that our design meets the following eligibility requirements (**review and check off all**):

- ☒ Be contained within a Pelican case as specified in Section 2.3 (except for laptops or tablets required for data collection and observation which can be mounted on top of the case). The case must remain closed during each parabola, but can be opened during level flight, if needed. There will also be a storage box available on board for storage of auxiliary equipment.
- ☒ Weight no more than 45 kg (not including the Pelican case).
- ☒ The experiment main power must be routed from the standard 115V/5A power supplied by the aircraft into the Pelican case power outlet.
- ☒ The sum of the maximum device currents contained in the experiment must not exceed the 5A rated current of the power source.
- ☒ Peak power consumption must be below 600 W.
- ☒ If applicable, use only dry cells, zinc-air, alkaline, or Ni-Cad batteries. Electrolyte or lithium-type batteries should be avoided whenever possible. When unavoidable, such as in commercial electronics, teams must be able to demonstrate batteries are in good health. It is best to purchase new brand name batteries; third party imitations should be avoided.
- ☒ The experiment must not contain free-floating items during the parabolas. All components must be tethered or contained within a secondary containment unit inside the Pelican case.
- ☐ Be free of materials classified as physical, health or environmental hazards under Canada's Hazardous Products Act such as high-pressure, toxic, corrosive, explosive and flammable materials. See Section 12.1. Note that any non-hazardous substance requiring secondary containment (e.g. water or dust) will need to be structurally tested for maximum loads experienced during the flight.
- ☒ Be free of hazardous radiation (e.g. Class 3 and 4 lasers).
- ☒ The experiment must fall to a safe configuration in the event of electrical power loss.
- ☒ Exclude the use of human and animal test subjects which would require Research Ethics Board approval. See note about biological specimens below.
- ☒ Not rely exclusively on automation to be actioned (e.g. using accelerometer data to begin a sequence). There must be a manual trigger to start your experiment.
- ☒ The experiment must demonstrate the ability to withstand 2g downwards loading with a safety factor of 1.5.



A minimum of two full-cycle ground tests of the experiment must be demonstrated prior to flight. The first demonstration will be submitted by video recording prior to integration. The second demonstration will be live at the campaign site at the time of integration.

If biological specimens are used, it must:

1. Not present any risk to experimenters, the flight crew and the aircraft. Specifically, the specimen must fall under **Biosafety Level/Risk Group 1**. You should submit a safety data sheet, or point to a database (e.g. <https://health.canada.ca/en/epathogen>) or other resource which shows the Risk Group (RG) classification of your specimen.
2. Be contained within a sealed space capable of withstanding the physical parameters of parabolic flight (g-forces, pressure, temperature, vibration). See Section 2.3.
3. Not be handled directly by experimenters during flight.

SECTION 3 – PROPOSAL

Section 1: Introduction

1.1 Research Motivation: Crystallization is a foundational process in materials science, with applications in pharmaceutical drug development¹⁻⁵, organic electronics⁶⁻⁸, energy storage^{9,10}, and semiconductor manufacturing¹¹. Interest in space-based crystallization processes is on the rise and has developed into a distinct part of the space industry¹²⁻¹⁶. A promising new ground-based technique involves laser-induced crystallization¹⁷, where short-lived bubbles rapidly reduce pressure and trigger nucleation. This method shortens crystal formation time from hours to seconds, with enhanced control over crystal size¹⁸, offering potential for high-throughput production of nano- and microcrystals¹⁸. However, the role of gravity in these dynamics, especially in laser-induced methods, remains poorly understood. Clarifying this influence is important for advancing space-based manufacturing and improving Earth-based simulations of space-grown materials.

This research aims to demonstrate laser-based crystallization in a microgravity environment and to compare the resulting crystal structures with those formed on Earth. The experiment will assess whether laser-induced crystallization under microgravity conditions accelerates crystal formation and enhances sample quality relative to Earth-grown crystals. These insights are particularly relevant for producing crystals suitable for diffraction-based structural determination using electron beams, which are now sufficiently compact and portable for spaceflight applications.

Key stakeholders include pharmaceutical companies that depend on high-quality crystals for drug development¹⁻⁵, Canadian startups looking for new niches in the space industry¹⁹, and the Canadian Space Agency, which supports research that strengthens Canada's role in the emerging space economy. Environmental considerations are also important. The proposed laser-based crystallization approach requires only short suborbital flights to produce thousands of samples, rather than relying on traditional orbital missions. This approach would reduce the number of satellites in low Earth orbit, mitigating space debris, require less fuel per flight, lower the overall environmental impact of space-based crystallization processes, and decrease operational costs.

The motivation behind this experiment is to lay the groundwork for scalable crystallization techniques in space, enabling space-based manufacturing, high-throughput screening and structural analysis using electron diffraction. By bridging fundamental science with engineering implementation, this research addresses critical questions about gravity's role in materials science.

1.2 Novelty of the Experiment: The advantage of crystal growth in microgravity has been discussed since the mid-1950s¹⁶ and has been demonstrated using orbital platforms^{1,2,16,20-23}. Meta-reviews widely agree that microgravity improves crystal size and quality^{16,20,23}. However, the primary barrier to commercialization and widespread adoption of microgravity-grown crystals is the high cost associated with using commercial space station platforms. This is where laser-induced methods become attractive: laser-based techniques can operate within sub-orbital trajectories, significantly reducing operational costs compared to orbital missions.

To enable this shift, initial experiments demonstrating laser-based crystallization in microgravity are needed. Notably, we found no flight-based microgravity crystallization experiments in the literature. This absence is likely due to the short duration of microgravity in sub-orbital flights, which is insufficient for crystal growth using conventional methods.

The proposed experiment introduces laser-based crystallization in a microgravity environment lasting only a few seconds, leveraging laser cavitation to produce crystals on sub-second timescales (dramatically faster than the hours to weeks required by traditional techniques²⁴). Because nucleation and growth occur so rapidly, laser-based methods could make sub-orbital flights a viable and cost-effective alternative to orbital platforms. In addition to speed, this technique offers precise control over nucleation and growth, making it highly suitable for high-throughput applications.

By combining laser technology with reduced gravity conditions, this research presents a novel approach to studying crystallization dynamics and contributes new insights to both fundamental science and applied materials engineering.

1.3 Goals: The goal of the proposed experiment is to demonstrate the production of microcrystals using laser cavitation-based crystallization in a short-duration microgravity environment (lasting less than 20 seconds), and to compare them to crystals produced using the same technique under Earth's gravity. This objective includes demonstrating that laser-induced crystallization can reliably occur in microgravity and determining whether gravity influences the morphology and structural characteristics of the resulting crystals.

1.4 Importance to Canada's Space Sector: Canada is a recognized leader in space robotics, satellite technology, and advanced materials research. This experiment supports Canada's strategic interest in space-based manufacturing and materials science by enabling crystallization processes compatible with short-duration microgravity environments, such as those found in sub-orbital flights. The global market value of space-based crystallization is estimated to be worth over \$450 million USD by the end of 2033²⁵. This project directly contributes to increasing accessibility to that market by demonstrating a rapid, cost-effective method for crystal production.

The research has direct commercial relevance for Canadian companies such as NordSpace, which could offer crystallization services on sub-orbital flights. These flights could be launched from Canada's Atlantic Spaceport Complex (ASX), positioning Canada as a competitive provider of space-based materials processing. A measurable impact of this project would be the enablement of 10 or more sub-orbital crystallization flights per year, creating new economic opportunities and reinforcing Canada's leadership in the emerging space economy.

1.5 Requirement for microgravity: The use of parabolic flight is essential because it provides a short-duration microgravity environment (approximately 20 seconds) that is sufficient for the rapid crystallization process triggered by laser cavitation.

Laser-induced crystallization occurs on sub-second timescales, with nucleation initiated by pressure drops from bubble collapse and crystal growth completing within seconds. These dynamics are ideally suited to the microgravity window offered by parabolic flight. The experiment will show that laser-based crystallization can reliably occur in reduced gravity, validating the method for future sub-orbital manufacturing applications.

Conducting this demonstration in microgravity eliminates gravitational interference, allowing for a direct comparison with Earth-based results. This will confirm whether the technique is compatible with short-duration spaceflight and whether microgravity enhances crystal quality or formation speed. The outcome will support the development of scalable, cost-effective crystallization platforms for Canadian space companies and contribute to Canada's leadership in space-based materials processing.

1.6 Research Hypothesis: Laser-induced crystallization can reliably occur in short-duration microgravity environments of less than 20 seconds, and the resulting microcrystals will exhibit distinct morphological and structural characteristics compared to those formed under Earth's gravity. Specifically, we predict that crystals formed in microgravity will be larger and contain fewer structural defects, as evidenced by improved diffraction quality, due to the absence of gravitational interference during nucleation and growth.

Section 2: Experimental Concept Design

2.1 Scientific Objectives: The primary scientific objective of this experiment is to investigate the role of gravity in laser-induced crystallization processes, specifically using laser cavitation-based crystallization. By conducting the experiment in a reduced gravity environment, we aim to isolate gravitational effects and determine whether they influence crystal formation, structure, or growth dynamics. To achieve this, we will quantify the impact of gravity on crystal morphology and growth kinetics by comparing crystals formed under microgravity and hyper gravity conditions using identical laser parameters and material conditions. The ideal sample for this experiment is a well studied system that is known to crystallize in microgravity using conventional methods, while also forming crystals under laser driven cavitation. An example of such a system includes hen egg white lysozyme^{26,27}.

2.2 Science Traceability Matrix

Scientific Objective A specific line of investigation that supports the overall Mission Goal	Scientific Measurement		Scientific Instrument	
	Measurement Objective Description of an observable physical phenomena that is key to understanding the Scientific Objective	Measurement Requirement Definition of the minimum measurement characteristics required to meet the Scientific Objective	Instrument Name of the tool that is required to conduct the measurement.	Instrument Requirement Definition of the minimum instrument performance metrics to meet the measurement requirement.
Quantification of crystal size	Determine if gravity changes the crystal size and shape.	Average and maximum Crystal size after formation in microgravity, hyper gravity, and normal gravitational conditions.	Electron microscope.	Electron microscope resolution below 1 nm. Field of view of 10 μm .
Quantification of crystalline structure and impurities	Determine if gravity changes the crystalline structure.	Crystal lattice structure, lattice spacing, and impurities.	Electron diffraction using electron microscope.	System needs to be able to resolve the structure at sub 0.2 nm resolution.
Quantification of system acceleration (Gravity)	Ensure crystal formation occurs at microgravity and hyper gravity conditions.	Measure acceleration of the container holding the solution.	Accelerometer	The instrument needs to be able to record $\pm 2\text{G}$, with a resolution of 0.1 G.
Mission Requirements Describe the scenario that must occur to result in a successful measurement for the mission.				
The mission specialist must conduct two distinct experimental phases: one under stable 2G hyper gravity conditions and another during microgravity. At the start of each phase, the laser cavitation system, housed within the interlocked chamber, is activated via the GUI on the control laptop to operate continuously for 10 seconds on a designated well of the 36-well plate, which is mounted on a motorized translation stage. After each trial, the stage automatically advances to a new well for duplicate trials, ensuring multiple independent samples for both gravity conditions without manual intervention. Throughout both phases, the accelerometer and camera continuously log synchronized data, with each well's position and corresponding gravity condition recorded for post-flight analysis. The GUI verifies interlock status and operational readiness before each laser activation, and a minimum of three replicates per gravity condition must be completed to ensure statistical validity and reproducibility. After crystal production in microgravity, the crystals will be harvested onto copper grids and stored at $-20\text{ }^{\circ}\text{C}$ until they can be analyzed using electron microscopy at the home university.				

2.3 System Architecture: To investigate the role of gravity in laser-based crystallization, the experiment integrates a compact laser cavitation system designed for safe and reliable operation in a microgravity environment (see Figure 1). The system consists of five main subsystems:

- (1) the laser cavitation chamber,
- (2) the fluid handling system,
- (3) crystallization solution and cavitation monitoring sensors,
- (4) the control and data acquisition system, and
- (5) the safety and structural enclosure.

Mechanical Mounting: The cavitation chamber assembly is rigidly mounted to the bottom panel of the Pelican case using vibration-damping brackets to minimize oscillations during parabolic maneuvers. Within this chamber:

- The laser module is bolted directly to the base plate of the chamber, aligned along the optical axis.
- A 20 mm CaF₂ focusing lens is mounted in a precision lens holder fixed to the same base plate, ensuring stable alignment with the laser beam path.
- The 36-well plate, containing the crystallization solution, is positioned above the laser beam path on a motorized translation stage (three-dimensional translation with 500 μm minimum step size). The stage is secured to the chamber floor using threaded fasteners and vibration-isolating spacers.
- The laser beam is focused through the lens onto the wells, with the optical path enclosed by a laser-safe housing that includes interlocked access doors. These interlocks ensure the laser can only operate when the housing is fully closed, maintaining Class 1 laser safety compliance.
- Inside the laser cavitation chamber, the CCD camera is fixed to an adjustable bracket attached to the chamber wall, providing a stable view of the well plate during crystallization.
- The accelerometer is mounted on a custom aluminum frame inside the Pelican case, mechanically isolated from the chamber to reduce noise from stage motion.

Electrical and Data Connections: Components draw power from the aircraft's 115 V AC, 5 A supply, routed through a power strip inside the case. The peak total system load is less than 600 W.

Laser Module: To drive laser cavitation, a Mid-IR laser with a pulse duration shorter than 10 ns, with a pulse energy of 10 μJ and repetition rate over 1 kHz is required. The laser also needs to withstand over 2G of acceleration. We will use a Glucoloop Ivy laser (operating manual attached), which is tested to meet the specified acceleration loads (3G). To focus the laser, a 20 mm CaF₂ focusing lens will be used. Bubble formation will be monitored with a camera and lens integrated into the laser module. In the absence of bubble formation, the well plate position will be adjusted to optimize focus, identified by the appearance of the largest bubble within the crystallization solution. To meet program requirements, the laser is enclosed in a laser safe box with an interlock, meeting Class 1 laser compliance.

Crystallization solution: Hen egg-white lysozyme (HEWL, Sigma-Aldrich, SDS Attached) will be used without further purification. Six stock solutions, with concentrations between 1 and 1000 mg in

10 mL of 0.1 M acetate buffer (pH 4.35), will be prepared based on previous literature and ground testing of the crystallization process. The selected concentrations will include one solution that does not crystallize under ground testing conditions, four that crystallize only under laser-induced conditions, and one that crystallizes without laser exposure. Each solution will be filtered through a 0.22 μm syringe filter to remove particulates and then pipetted into the wells of a 36-well plate, completely filling each well. To prevent evaporation, each well will be sealed with a sterile glass coverslip.

Control Interfaces: The laser and translation stages are connected to an intermediate controller via RS-232 serial connections (DB9 connectors). The intermediate controller communicates with the control laptop via USB 3.0, enabling command execution and status feedback.

Sensor Data Interfaces: The camera streams video directly to the laptop via USB 3.0. The accelerometer transmits real-time acceleration data to the laptop via USB or RS-485, depending on the sensor configuration.

Cable Management: All cables are routed through grommets pass-throughs in the Pelican case with strain relief clamps to prevent disconnection during high-g maneuvers.

Control and User Interface: The mission specialist operates the system through a custom GUI on the control laptop. The GUI integrates all functions, including laser firing, stage positioning, and real-time monitoring of camera and accelerometer data. The laptop supervises the intermediate controller for laser and stage operations while directly logging sensor data. Prior to each run, the GUI verifies that all interlocks are engaged and that the gravity conditions meet the experiment's requirements. Data from the accelerometer and camera are time-synchronized to allow accurate correlation with crystallization events.

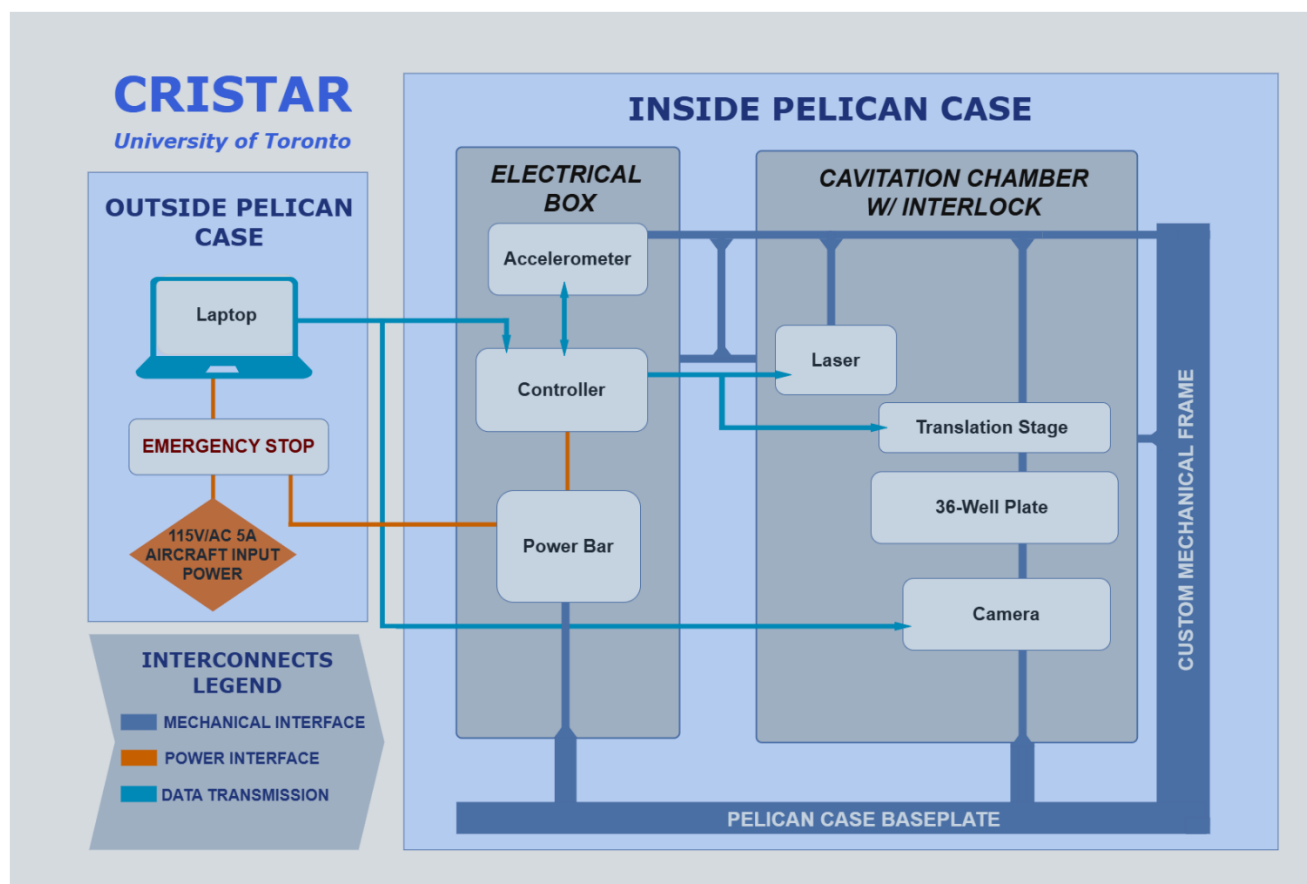


Figure 1: Block Diagram of the system architecture used to produce crystals using laser-induced crystallization.

Section 3: Concept of Operations

3.1 Equipment Requirements

Onboard the Aircraft: The experiment consists of a compact laser cavitation system integrated into a Pelican case for secure operation in a microgravity environment. This system requires the use of a short pulse laser with a peak intensity high enough to drive cavitation. A mid- IR laser with a pulse energy higher than 10 μJ is sufficient if focused to 10 microns. One such laser that is suitable for this is a Mid-IR Ivy Glucoloop laser (which meets the specified acceleration loads), with a 20 mm CaF_2 focusing lens. For the super saturated solution where the cavitation bubble will drive crystal formation a 36-well plate will be completely filled with solution and mounted on a motorized translation stage inside a laser-safe enclosure with interlocks for Class 1 compliance. The laser beam is aligned to focus on individual wells, and the stage automatically advances between wells for sequential trials under different gravity conditions.

System control is managed through a custom GUI on a laptop, which communicates with an intermediate controller via USB. The controller executes serial commands to operate the laser and translation stage, while the camera and accelerometer send real-time data directly to the laptop for monitoring and logging. All components draw power from the aircraft's 115 V AC, 5 A supply, with total consumption under 600 W.

The Pelican case houses all hardware on a custom internal frame for mechanical stability, with vibration-damping mounts and strain-relieved cable routing to ensure reliability during parabolic maneuvers. Safety features include interlocked laser housing and GUI-based system checks before activation.

At the Home University: Facilities are required for machining and assembling the custom internal frame and Pelican case system, as well as for precise alignment of the laser, focusing optics, and translation stage. These integration steps will be completed in the physics and engineering machine shops on the university campus. Post-flight analysis will utilize advanced characterization tools, including electron microscopes with diffraction capability for structural determination of nano-crystals. These systems are accessible at the University of Toronto's Centre for Nanostructure Imaging, providing high-resolution imaging and crystallographic analysis. The specific system to be used for this work is a Hitachi HT7700 TEM which has better than 1.5 nm resolution electron microscope imaging and 2 angstrom resolution diffraction resolution for measuring the crystal structure. Additional resources include spectroscopy and microscopy facilities for detailed evaluation of crystallized samples, along with videography equipment for documentation during integration and testing.

3.2 Environmental Requirements: The experimental apparatus will be exposed to several environmental conditions during integration and flight that could pose risks to achieving the scientific objectives. These include temperature variations, mechanical loads, vibration, and acceleration changes during parabolic maneuvers.

Temperature: Ambient temperatures at the integration site and onboard the aircraft are expected to range between 15°C and 30°C, which is within the operating limits of the Ivy Glucoloop laser and associated electronics. However, sudden temperature fluctuations could affect laser alignment and

sensor calibration. To mitigate this, all components will be housed in an insulated Pelican case to minimize thermal gradients, and sensors will be pre-calibrated at expected operating temperatures.

Mechanical Loads: During handling and installation, the system may experience static loads and shocks that could damage optical alignment or structural integrity. To address this, the laser chamber, translation stage, and electrical components will be mounted on a custom internal frame secured to the Pelican case base plate, ensuring structural rigidity and even load distribution. All fasteners will use vibration-resistant hardware to prevent loosening during transport and flight.

Vibration and Acceleration: Parabolic flight introduces transient vibrations and accelerations during pull-up and recovery phases, which could disrupt optical stability or stage positioning. To mitigate these effects, critical components such as the laser and translation stage will be mounted with vibration-damping pads, and optical elements will be locked in precision mounts. Accelerometer data will be used to confirm valid microgravity periods for data collection, ensuring that only measurements taken under stable conditions are analyzed.

Radiation and Electromagnetic Interference: While no significant radiation hazards are anticipated during the flight, electronic components will be shielded within the Pelican case to minimize electromagnetic interference that could affect data integrity.

Operational Limits: The system will operate within a controlled temperature range of 15°C to 30°C, with storage maintained at 20°C $\pm$ 5°C. Fluid handling and laser operation will occur only during verified microgravity periods, as indicated by accelerometer readings, to ensure experimental validity.

3.3 In-flight Operations:

To test the six crystallization solution conditions with three replicates each, a minimum of 16 parabolas is required. Between flights, the 36-well plate will be replaced with a fresh one. The inflight operation is described in detail below.

Level Flight Prior to Parabola: Before the aircraft enters the parabolic maneuver, the mission specialist will verify that the experimental system is powered, all safety interlocks are engaged, and the control laptop with GUI is operational. The accelerometer and camera will be checked for stable readings within expected parameters. The translation stage will be homed, and the first well in the 36-well plate will be positioned under the laser focal point. Note: the stage will not be moved in 2G flight to prevent damage to the system. Once these checks are complete, the mission specialist will stand by for the command to begin the experiment during the 2G phase.

Initial 2G Phase: During the initial 2G phase, the mission specialist will initiate the first experiment under hyper gravity by activating the laser through the GUI manually. The laser will operate for 10 seconds. Data from the accelerometer and camera will be logged continuously to correlate crystallization events with gravity conditions.

Microgravity Phase: As soon as microgravity is reached the mission specialist will move to a new well and will again operate for 10 seconds. After the 10 seconds while still in microgravity the stage

will advance to the next well position. This process ensures that both gravity conditions (2G and microgravity) are tested within the same parabola when possible.

Recovery 2G Phase: Before exiting microgravity, the mission specialist will ensure the laser is off and that the stage is not moving. All data will be saved automatically through a live recording file. This sequence will be repeated until all required replicates for each gravity condition are completed.

Post Flight: Following flight operations, crystals formed in the 36-well plate will be carefully extracted and filtered from the liquid solution onto TEM grids for structural analysis at the home university. The selected samples are inherently temperature stable, so no special thermal storage or transport conditions are required. Post-flight characterization will include high-resolution electron microscopy with diffraction capability to determine crystal morphology and lattice structure, ensuring accurate assessment of the effects of gravity on crystallization.

Section 4: Project Plan

4.1 Funding Strategy: To ensure the success of the proposed experiment, a multi-tiered funding and resource strategy has been developed. This strategy leverages institutional support, internal departmental resources, corporate sponsorships, and student grants, with specific team members assigned to manage each component.

Core Equipment Access: The Ivy Glucoloop Mid-IR laser and associated optical components will be provided by the faculty advisor. This confirmed support ensures that the core experimental apparatus is available at no additional cost and without eligibility constraints. The Team Lead is responsible for coordinating this access and maintaining communication with the advisor throughout the project.

Material and Chemical Resources: Materials required for system assembly and integration will be sourced from surplus departmental supplies under the supervision of the faculty advisor. Chemicals necessary for crystallization will be provided by the advisor and supplemented through collaboration with researchers in the Department of Biochemistry at the University of Toronto. These resources are secured through internal agreements and do not require external funding. The Science Lead will oversee the acquisition and ensure compatibility with experimental protocols.

Institutional Imaging Support: High-resolution imaging of crystallized samples will be conducted at the Centre for Nanostructure Imaging (CNI) at the University of Toronto. This facility offers access to advanced nano-imaging tools such as Scanning Electron Microscopes (SEMs) and transmission electron microscopes (TEMs). The Outreach Coordinator is responsible for engaging with CNI staff to secure access, manage scheduling, and ensure compliance with facility requirements.

Industry Sponsorships: To enhance system capabilities, the team is pursuing sponsorships with key industry partners. Edmund Optics is being approached for high-resolution cameras to support optical diagnostics, with the Electrical Lead managing technical specifications and sponsorship. Thorlabs is being contacted for optical mounting hardware essential for system integration, with the Structural

Lead responsible for securing components and ensuring mechanical compatibility. These sponsorships are currently under discussion and contingent on formal agreements.

Student and community funds: The team will apply for a Special Activity Grant from the SPIE Student Chapter, which typically provides up to \$1000 CAD per project. The Outreach Lead will coordinate this process, ensuring the application is submitted at least three months in advance of funding needs and aligns with the promotion of optics and science. These funds will be dedicated to the outreach portion of this proposal.

Additional institutional support will be sought from the IEEE Student Chapter, which offers access to electronics and laboratory infrastructure. Based on existing memberships, the Structural Lead and Science Lead will jointly manage engagement with IEEE, ensuring eligibility and coordinating access to relevant resources.

The Outreach Coordinator will also apply to the Photonics Innovation Center at the University of Toronto for hardware support, including electronics and integration components relevant to the optical system.

Finally, the Team Lead will manage applications to the Massey College Junior Fellow Initiative, which provides \$100 CAD per application to offset transportation costs. These small grants are available year-round and require active student membership for consideration.

4.2 Outreach Strategy: Our outreach strategy includes a series of diverse activities designed to engage the general public, K–12 students, and the academic community, while demonstrating the real-world impact of space research and crystallization in microgravity.

To engage the general public, the team will participate in Science Rendezvous at the University of Toronto in May 2026. This nationally recognized science festival provides an opportunity to showcase our experiment and its relevance to space-based materials science. The Outreach Lead will coordinate booth setup, interactive demonstrations, and public engagement activities.

For K–12 students, we will conduct both virtual and in-person outreach sessions targeting rural and remote communities. Confirmed events include:

- Hazelton, Terrace, and Smithers, BC – January 2026, as part of an ongoing effort with the Photonics Innovation Center
- High schools in Durham and Peel Region – Spring 2026
- Mississaugas of the Credit First Nation – Spring 2026, in collaboration with Massey College
- University of Toronto’s DEEP Summer Academy – Summer 2026
- High schools in Toronto – Fall 2026
- Additional sessions across underrepresented communities in Ontario and Canada – Fall 2026

These sessions will focus on students in grades 6–12, aiming to inspire interest in STEM and space exploration. The Outreach Lead will manage logistics and content delivery, with support from other team members for hands-on activities and Q&A sessions.

To engage the academic community, we plan to deliver a colloquium talk to the Physics and Chemistry Departments at the University of Toronto. This event will be facilitated by the graduate student members of the team and will focus on presenting experimental results, discussing the implications of microgravity crystallization, and fostering interdisciplinary dialogue.

To reach broader audiences, we will produce a YouTube documentary chronicling the journey from concept to flight. This will be complemented by active engagement on social media platforms (Instagram, X, LinkedIn), where we will share updates, behind-the-scenes content, and educational insights. The Team Lead will oversee content strategy and analytics tracking.

At all outreach events, we will present:

- Structural comparisons of crystals formed under microgravity vs. Earth gravity
- Data visualizations of crystal size and yield
- References to previous ISS-based studies

These demonstrations will illustrate how big-picture scientific questions can be transformed into practical engineering solutions, reinforcing the connection between theory and application.

Measurable outreach goals include:

- Event attendance (target: 500+ across all events)
- Social media engagement (target: 10,000+ followers, 10,000+ views, 500+ shares)
- Survey responses assessing knowledge gained and interest in STEM careers (target: 100+ responses with >80% positive impact)

5.0 References:

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SECTION 4 – SUMMARY AND SIGNATURES

Submit a **single PDF file** containing the following items to canrgx@seds.ca by **11:59 PM EST on Sunday, October 19, 2025.**:

- ☒ Completed and signed application form
- ☒ Completed proposal
- ☒ Proof of enrollment for each team member

Please name the PDF file using the following format: Teamname_RGX_Application_year

Team Lead

- ☒ **Timeline Acknowledgement:** I certify that the team will adhere to the milestone timeline of the CAN-RGX competition. I certify that it is the responsibility of the Team Lead to monitor the CAN-RGX website (seds.ca/can-rgx) and monitor communications from SEDS-Canada for changes to the timeline.
- ☒ **Funding Expectations:** I acknowledge that team is responsible to fully fund the development of our project and the logistics of travel to the Campaign site.
- ☒ **Publicity Guidelines:** I shall acknowledge SEDS-Canada in the organization of this campaign and include SEDS-Canada as a contributor in external publicity, social media materials, outreach activities.
- ☒ I certify that I will serve as a team lead and a primary point of contact for the team. I agree to inform SEDS-Canada at canrgx@seds.ca promptly of any changes to the team lead position and primary point of contact.

Signature: 

Date: October 16 2025

Faculty Advisor

- ☒ I certify that I will serve as a Faculty Advisor for this project, and I understand that I will be asked to provide guidance and support through some or all of the phases of the challenge, including submission of the project Proposal, Preliminary Design Review, Critical Design Review, and other milestones.

Signature: 

Date: October 16, 2025



May 16, 2025

To Whom It May Concern:

This letter is to confirm that Alexander Wainwright (Student# 1005039343) is for the 2024-2025 academic year (September 1, 2025 to August 31, 2026) at the School of Graduate Studies, University of Toronto. The registration record is as follows:

Faculty:	Faculty of Arts and Science
Department:	Department of Physics
Program of Study:	PhD in Physics
Type of Program:	Doctoral Program
Registration Status:	Full-time
Program Start Date:	Fall 2023
Supervisor:	R. J. Dwayne Miller

Alexander Wainwright started the PhD in Physics degree program in Fall 2023.

Please do not hesitate to contact our office at graduate.information@utoronto.ca should you need further information.

Sincerely,

A handwritten signature in black ink, appearing to read "Bd", with a stylized flourish extending to the right.

Brian Desrosiers-Tam, Assistant Dean
School of Graduate Studies
University of Toronto
63 St. George Street
Toronto, Ontario Canada M5S 2Z9
Tel: 416-978-6614 Email: graduate.information@utoronto.ca
www.sgs.utoronto.ca

This letter is electronically generated. The information it contains reflects the information in the University of Toronto's Student Information System on the date and at the time when the letter was generated.



May 26, 2025

To Whom It May Concern:

This letter is to confirm that Alexander Wainwright (Student# 1005039343) is admitted for the 2025-2026 academic year (September 1, 2025 to August 31, 2026) at the School of Graduate Studies, University of Toronto. The registration record is as follows:

Alexander Wainwright started the PhD in Physics degree program in Fall 2023.

Please do not hesitate to contact our office at graduate.information@utoronto.ca should you need further information.

Sincerely,

A handwritten signature in black ink, appearing to read "Bd", with a long, sweeping horizontal stroke extending to the right.

Brian Desrosiers-Tam, Assistant Dean
School of Graduate Studies
University of Toronto
63 St. George Street
Toronto, Ontario Canada M5S 2Z9
Tel: 416-978-6614 Email: graduate.information@utoronto.ca
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This letter is electronically generated. The information it contains reflects the information in the University of Toronto's Student Information System on the date and at the time when the letter was generated.



UNIVERSITY OF TORONTO
FACULTY OF APPLIED SCIENCE & ENGINEERING

The information contained in this document is accurate as of October 15, 2025.

Student Name Altomare, Lauren Canton		Student Number 1008998649	
Student Address 265 Poplar Plains Road, Toronto, ON M4V2N9 Canada			
Degree Program Faculty of Applied Science & Engineering, Engineering Science (Biomedical Systems)			
Current Academic Session Fall 2025		Academic Session Start Date September 1, 2025	
Year of Study Year 4	Registration Status Registered	Attendance Full-time	PEY Co-op No

Notes:

- The Engineering degree program is a four-year year program with an optional additional year if the student participates in the PEY Co-op program.
- In the Faculty of Applied Science & Engineering, "year of study" reflects satisfactory completion of the pre- requisite year.
- University of Toronto students in the Faculty of Applied Science & Engineering are considered registered only after they have selected courses and paid fees.
- The DLI number for the University of Toronto is O19332746152.



UNIVERSITY OF TORONTO
FACULTY OF APPLIED SCIENCE & ENGINEERING

The information contained in this document is accurate as of October 19, 2025.

Student Name Barsa, Elias Alexander Gabriel		Student Number 1009943518	
Student Address 57 Charles St West, Toronto, ON M5S2X1 Canada			
Degree Program Faculty of Applied Science & Engineering, Engineering Science (Physics)			
Current Academic Session Fall 2025		Academic Session Start Date September 1, 2025	
Year of Study Year 3	Registration Status Registered	Attendance Full-time	PEY Co-op No

Notes:

- The Engineering degree program is a four-year year program with an optional additional year if the student participates in the PEY Co-op program.
- In the Faculty of Applied Science & Engineering, "year of study" reflects satisfactory completion of the pre- requisite year.
- University of Toronto students in the Faculty of Applied Science & Engineering are considered registered only after they have selected courses and paid fees.
- The DLI number for the University of Toronto is O19332746152.



UNIVERSITY OF TORONTO
FACULTY OF APPLIED SCIENCE & ENGINEERING

The information contained in this document is accurate as of October 19, 2025.

Student Name Caribe, Gabriel Hunter		Student Number 1010376932	
Student Address 55 Barrydale Court, London, ON N6G4G4 Canada			
Degree Program Faculty of Applied Science & Engineering, Engineering Science (Physics)			
Current Academic Session Fall 2025		Academic Session Start Date September 1, 2025	
Year of Study Year 3	Registration Status Registered	Attendance Full-time	PEY Co-op Yes

Notes:

- The Engineering degree program is a four-year year program with an optional additional year if the student participates in the PEY Co-op program.
- In the Faculty of Applied Science & Engineering, "year of study" reflects satisfactory completion of the pre- requisite year.
- University of Toronto students in the Faculty of Applied Science & Engineering are considered registered only after they have selected courses and paid fees.
- The DLI number for the University of Toronto is O19332746152.



UNIVERSITY OF TORONTO
FACULTY OF APPLIED SCIENCE & ENGINEERING

The information contained in this document is accurate as of October 19, 2025.

Student Name Mahdia, Syeda Nusaiba		Student Number 1010244576	
Student Address 19 Montvale Drive, Scarborough, ON M1M3E5 Canada			
Degree Program Faculty of Applied Science & Engineering, Engineering Science (Physics)			
Current Academic Session Fall 2025		Academic Session Start Date September 1, 2025	
Year of Study Year 3	Registration Status Registered	Attendance Full-time	PEY Co-op Yes

Notes:

- The Engineering degree program is a four-year year program with an optional additional year if the student participates in the PEY Co-op program.
- In the Faculty of Applied Science & Engineering, "year of study" reflects satisfactory completion of the pre- requisite year.
- University of Toronto students in the Faculty of Applied Science & Engineering are considered registered only after they have selected courses and paid fees.
- The DLI number for the University of Toronto is O19332746152.



UNIVERSITY OF
TORONTO

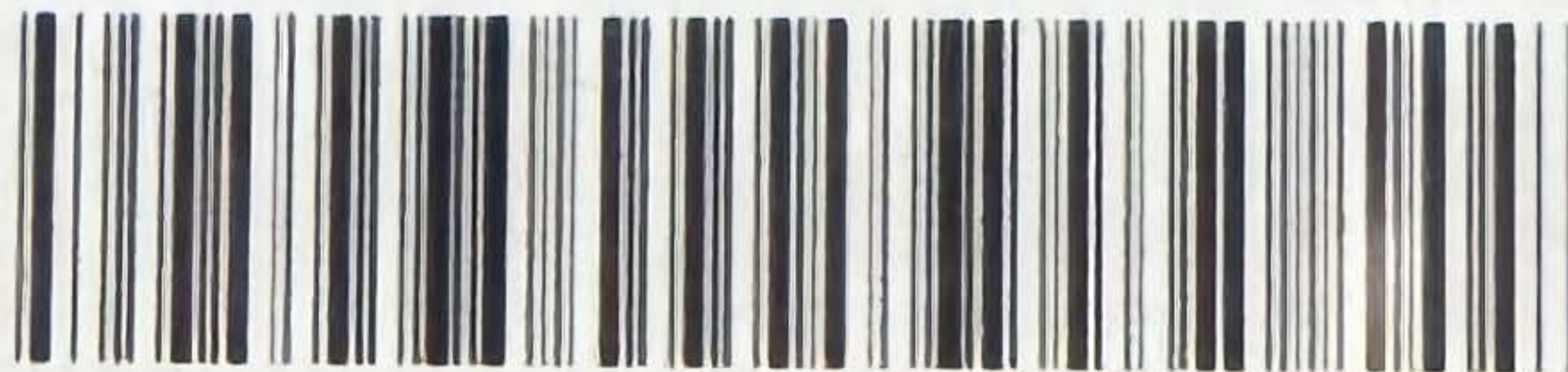
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Issued:

31-August-2021

• SAFETY DATA SHEET

Version 6.6
Revision Date 15.09.2025
Print Date 16.09.2025

SECTION 1. IDENTIFICATION

1.1 Product identifiers

Product name : Lysozyme, from chicken egg white

Product Number : L3790

Brand : Sigma

CAS-No. : 12650-88-3

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses : Laboratory chemicals, Synthesis of substances

1.3 Details of the supplier of the safety data sheet

Company : MilliporeSigma Canada Ltd.
2149 WINSTON PARK DRIVE
OAKVILLE ON L6H 6J8
CANADA

Telephone : +1 905 829-9500

Fax : +1 905 829-9292

1.4 Emergency telephone number

Emergency Phone # : +1-703-527-3887 CHEMTREC
(International)
24 Hours/day; 7 Days/week

SECTION 2. HAZARDS IDENTIFICATION

GHS classification in accordance with the Hazardous Products Regulations

Respiratory sensitisation : Category 1

GHS label elements

Hazard pictograms :



Signal Word : Danger

Hazard Statements : H334 May cause allergy or asthma symptoms or

breathing difficulties if inhaled.

Precautionary statements :

Prevention:

P261 Avoid breathing mist or vapours.

P284 Wear respiratory protection.

Response:

P304 + P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.

P342 + P311 If experiencing respiratory symptoms: Call a POISON CENTER/ doctor.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards

None known.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Concentration (% w/w)
glycerine	56-81-5	$\geq 30 - < 60$ *
Lysozyme	12650-88-3	$\geq 1 - < 5$ *

* Actual concentration or concentration range is withheld as a trade secret

SECTION 4. FIRST AID MEASURES

- General advice : First aiders need to protect themselves.
Show this safety data sheet to the doctor in attendance.
- If inhaled : After inhalation: fresh air. Call in physician.
- In case of skin contact : In case of skin contact: Take off immediately all contaminated clothing. Rinse skin with water/ shower.
- In case of eye contact : After eye contact: rinse out with plenty of water.
Remove contact lenses.
- If swallowed : After swallowing: immediately make victim drink water (two glasses at most).
Consult a physician.
- Most important symptoms and effects, both acute and delayed : The most important known symptoms and effects are described in the labelling (see section 2.2) and/or in section 11

Protection of first-aiders : For personal protection see section 8.

Notes to physician : No data available

SECTION 5. FIREFIGHTING MEASURES

Suitable extinguishing media : Foam
Carbon dioxide (CO₂)
Dry powder

Unsuitable extinguishing media : For this substance/mixture no limitations of extinguishing agents are given.

Specific hazards during fire fighting : Mixture with combustible ingredients.

Development of hazardous combustion gases or vapours possible in the event of fire.

Hazardous combustion products : Carbon oxides

Specific extinguishing methods : No data available

Further information : Prevent fire extinguishing water from contaminating surface water or the ground water system.

Special protective equipment for fire-fighters : In the event of fire, wear self-contained breathing apparatus.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures : Advice for non-emergency personnel:
Do not breathe vapours, aerosols.
Avoid substance contact.
Ensure adequate ventilation.
Evacuate the danger area, observe emergency procedures, consult an expert.
Advice for emergency responders:
For personal protection see section 8.

Environmental precautions	: Do not let product enter drains.
Methods and materials for containment and cleaning up	: Cover drains. Collect, bind, and pump off spills. Observe possible material restrictions (see sections 7 and 10). Take up carefully with liquid-absorbent material (e.g. Chemizorb®). Dispose of properly. Clean up affected area.

SECTION 7. HANDLING AND STORAGE

For precautions see section 2.2.

Advice on safe handling	: Work under hood. Do not inhale substance/mixture. Avoid generation of vapours/aerosols.
Further information on storage conditions	: Tightly closed. Keep locked up or in an area accessible only to qualified or authorised persons.
Storage class	: 10, Combustible liquids
Recommended storage temperature	: -20 °C

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
glycerine	56-81-5	TWA (Mist)	10 mg/m ³	CA AB OEL
		TWA (Mist)	10 mg/m ³	CA BC OEL
		TWA (Respirable mist)	3 mg/m ³	CA BC OEL
		TWAEV (Mist)	10 mg/m ³	CA QC OEL

Engineering measures : No data available

Personal protective equipment

Respiratory protection : required when vapours/aerosols are generated.

Our recommendations on filtering respiratory protection are based on the following standards: DIN

EN 143, DIN 14387 and other accompanying standards relating to the used respiratory protection system.

Recommended Filter type: : Filter type ABEK

The entrepreneur has to ensure that maintenance, cleaning and testing of respiratory protective devices are carried out according to the instructions of the producer. These measures have to be properly documented.

Hand protection

Material : Nitrile rubber
Break through time : 480 min
Glove thickness : 0.11 mm
Protective index : Full contact
Manufacturer : Dermatril® (KCL 740 / Aldrich Z677272, Size M)

Material : Nitrile rubber
Break through time : 480 min
Glove thickness : 0.11 mm
Protective index : Splash contact
Manufacturer : Dermatril® (KCL 740 / Aldrich Z677272, Size M)

Manufacturer : data source: KCL GmbH, D-36124 Eichenzell, phone +49 (0)6659 87300, e-mail sales@kcl.de, test method: EN374

Remarks : Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands.
If used in solution, or mixed with other substances, and under conditions which differ from EN 374, contact the supplier of the CE approved gloves. This recommendation is advisory only and must be evaluated by an industrial hygienist and safety officer familiar with the specific situation of anticipated use by our customers. It should not be construed as offering an approval for any specific use scenario.

Eye protection : Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).
Safety glasses

Skin and body protection : protective clothing

Hygiene measures : Change contaminated clothing. Preventive skin protection recommended. Wash hands after working with substance.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance : liquid

Color : No data available

Odor : No data available

Odor Threshold : No data available
pH : No data available

Melting point : No data available

Boiling point/boiling range : No data available

Flash point : No data available

Evaporation rate : No data available

Flammability (solid, gas) : No data available

Flammability (liquids) : No data available

Burning rate : No data available

Upper explosion limit /
Upper flammability limit : No data available

Lower explosion limit /
Lower flammability limit : No data available

Vapor pressure : No data available

Relative vapour density : No data available

Relative density : No data available

Density : No data available

Water solubility : No data available

Partition coefficient: n-
octanol/water : No data available

Autoignition temperature	: No data available
Decomposition temperature	: No data available
Viscosity, dynamic	: No data available
Viscosity, kinematic	: No data available
Flow time	: No data available
Explosive properties	: Not classified as explosive.
Oxidizing properties	: none
Particle characteristics	
Particle size	: No data available

SECTION 10. STABILITY AND REACTIVITY

Reactivity	: No data available
Chemical stability	: The product is chemically stable under standard ambient conditions (room temperature) .
Possibility of hazardous reactions	: No data available
Conditions to avoid	: no information available
Incompatible materials	: Strong bases Strong oxidizing agents
Hazardous decomposition products	: In the event of fire: see section 5

SECTION 11. TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Mixture

Acute toxicity

Oral: No data available

Inhalation: No data available

Dermal: No data available

Skin corrosion/irritation

No data available

Serious eye damage/eye irritation

No data available

Sigma - L3790

Page 7 of 13

The life science business of Merck KGaA, Darmstadt, Germany operates as MilliporeSigma in the US and Canada

**MILLIPORE
SIGMA**

Respiratory or skin sensitization

Mixture may cause allergy or asthma symptoms or breathing difficulties if inhaled.

Germ cell mutagenicity

No data available

Carcinogenicity

No data available

Reproductive toxicity

No data available

Specific target organ toxicity - single exposure

No data available

Specific target organ toxicity - repeated exposure

No data available

Aspiration hazard

No data available

11.2 Additional Information

prolonged or repeated exposure can cause:, Nausea, Dizziness

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

Components**glycerine****Acute toxicity**

LD50 Oral - Rat - female - 27,200 mg/kg

Remarks: (ECHA)

LC50 Inhalation - Rat - male and female - 4 h - > 5,850 mg/l - aerosol

Remarks: (ECHA)

LD50 Dermal - Guinea pig - male and female - 56,750 mg/kg

Remarks: (ECHA)

Skin corrosion/irritation

Skin - Rabbit

Result: No skin irritation - 24 h

Remarks: (ECHA)

Serious eye damage/eye irritation

Eyes - Rabbit

Result: No eye irritation - 7 Days

Remarks: (ECHA)

Respiratory or skin sensitization

Local lymph node assay (LLNA) - Mouse

Result: negative

(OECD Test Guideline 429)

Germ cell mutagenicity

Test Type: Ames test

Test system: S. typhimurium

Result: negative

Remarks: (IUCLID)

Test Type: In vitro mammalian cell gene mutation test

Test system: Chinese hamster ovary cells

Result: negative

Test Type: sister chromatid exchange assay

Test system: Chinese hamster ovary cells

Result: negative

Test Type: unscheduled DNA synthesis assay

Test system: rat hepatocytes

Result: negative

Test Type: Chromosome aberration test in vitro

Test system: Chinese hamster ovary cells

Result: negative

Carcinogenicity

No data available

Reproductive toxicity

No data available

Specific target organ toxicity - single exposure

No data available

Specific target organ toxicity - repeated exposure

Aspiration hazard

No data available

Lysozyme

Acute toxicity

LD50 Oral - Rat - male and female - > 5,050 mg/kg

(US-EPA)

LC50 Inhalation - Rat - male and female - 4 h - > 2.51 mg/l - dust/mist

(US-EPA)

LD50 Dermal - Rabbit - male and female - > 5,050 mg/kg

(US-EPA)

No data available

Skin corrosion/irritation

No data available

Serious eye damage/eye irritation

No data available

Respiratory or skin sensitization

Prolonged or repeated exposure may cause allergic reactions in certain sensitive individuals.

Germ cell mutagenicity

No data available

Carcinogenicity

No data available

Reproductive toxicity

No data available

Specific target organ toxicity - single exposure

No data available

Specific target organ toxicity - repeated exposure

No data available

Aspiration hazard

No data available

SECTION 12. ECOLOGICAL INFORMATION**Ecotoxicity****Components:****glycerine:**

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 54,000 mg/l
End point: mortality
Exposure time: 96 h
Test Type: static test
Remarks: (ECHA)

Persistence and degradability**Components:****glycerine:**

Biodegradability : aerobic
Inoculum: activated sludge
Concentration: 226 mg/l
Result: Readily biodegradable.
Biodegradation: 94 %
Exposure time: 1 d
Remarks: (ECHA)

Biochemical Oxygen Demand (BOD) : 870 mg/g
Incubation time: 5 d
Remarks: (External MSDS)

Chemical Oxygen Demand (COD) : 1,160 mg/g
Remarks: (External MSDS)

ThOD : 1,217 mg/g
Remarks: (Lit.)

BOD/ThOD : 71 %
Remarks: (Lit.)

Bioaccumulative potential

Components:

glycerine:

Partition coefficient: n-octanol/water : log Pow: -1.75 (25 °C)
pH: 7.4
Method: OECD Test Guideline 107
Remarks: Bioaccumulation is not expected.

Mobility in soil

No data available

Other adverse effects

No data available

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : Waste material must be disposed of in accordance with the national and local regulations. Leave chemicals in original containers. No mixing with other waste. Handle uncleaned containers like the product itself.

SECTION 14. TRANSPORT INFORMATION

International Regulations

IATA-DGR

Not regulated as a dangerous good

IMDG-Code

Not regulated as a dangerous good

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

National Regulations

TDG

Not regulated as a dangerous good

Special precautions for user

Remarks : Not classified as dangerous in the meaning of transport regulations.

SECTION 15. REGULATORY INFORMATION

Canadian lists

No substances are subject to a Significant New Activity Notification.

SECTION 16. OTHER INFORMATION

Full text of other abbreviations

CA AB OEL	:	Canada. Alberta, Occupational Health and Safety Code (table 2: OEL)
CA BC OEL	:	Canada. British Columbia OEL
CA QC OEL	:	Québec. Regulation respecting occupational health and safety, Schedule 1, Part 1: Permissible exposure values for airborne contaminants
CA AB OEL / TWA	:	8-hour Occupational exposure limit
CA BC OEL / TWA	:	8-hour time weighted average
CA QC OEL / TWA EV	:	Time-weighted average exposure value

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonised System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organisation; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardisation; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; MERCOSUR - The Agreement for the Facilitation of the Transport of Dangerous Goods; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organisation for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

The information is believed to be correct but is not exhaustive and will be used solely as a guideline, which is based on current knowledge of the chemical substance or mixture and is applicable to appropriate safety precautions for the product. It does not represent any

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CA / EN

	Doc No: PRD-ML-001
	Version No: 01
	Effective Date: 27/02/2024
USER MANUAL OF IVY LASER SERIES	

Operation manual



Nanosecond pulse mid-infrared laser

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1. For the quick user who does not want to read the manual

- The laser contains electro-static discharge (ESD) sensitive components. Use normal ESD protection precautions when handling the laser in order to avoid permanent damage. Always connect the earth point (see Chapter 7) to the earth of your facility or device.
- Do not attempt to open the laser head or laser driver. The laser is equipped with a tamper mechanism, which will detect opening of the laser. Warranty will be void and it will lead to permanent damage.

Quick start guide for users that control the laser with a computer:

- 1) Place the laser where you want to operate it, fix it with the three through-hole screws or clamp it at the screw positions in its place. Make sure the output beam area is well secured and that there are no significant back-reflections into the laser.
- 2) If a purging unit is used, please make sure to connect the filtered clean air output of the purger to the air inlet of the laser, and the air outlet of the laser to the air intake of the purger. Turn on the purger and let dry clean air circulate in the system.
- 3) Power the laser with either 24V of your system or with the supplied power supply.
- 4) Close the interlock loop.

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- 5) Connect the laser via Ethernet or USB or RS232 to a computer.
- 6) Start the graphical user interface of the Ivy laser.
- 7) In the “Communication” tab, select the mode of communication that you are using. Then press the corresponding “Refresh” button.
- 8) Identify the Ivy that you want to connect to in the pull down menu and select it and press the “Connect” button. In the bottom of the window it should now state “Connected”.
- 9) In the “Main” tab you can now set the laser to standby, which will initiate thermalization of the laser. Depending on the ambient temperature, this can take up to 10 minutes. A spinning wheel is displayed in the status area as long as thermalization is taking place.
- 10) Toggle the key switch.
- 11) Select if you want to use the internal clock or triggering the laser with an external clock and select the frequency you want to operate at.
- 12) Once the spinning wheel in the status area of GUI has disappeared, you can turn on laser emission by pressing the “Emission” button. “Emission on” and an emission symbol will be shown in the status area.
- 13) To turn off laser emission, press the “Standby” button, and the laser will go to standby. Thermalization will be maintained as long as the “Off” button is not hit.
- 14) For best stability, we recommend to keep the laser with standby or emission on 24/7/365. There are no consumables in the laser, and power consumption is low.

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2. Introduction

This manual contains information for operating Glucoloop's Ivy laser. The Ivy is an industrial all-in-one, turn-key, maintenance free, medium energy, high peak power, nanosecond laser source for a wide range of applications. The system design allows for stable and turn-key performance.

The laser system is build according to following standards:

- ANSI / ESD S20.20
- IPC-A-610 class 2
- Software documentation IEC 62304 class A

Each laser system undergoes strict highly accelerated stress screening (HASS) and lifetime testing procedures (HALT). Type tests according to ISO 9022 30-05-2, ISO 9022 10-02-1 and ISO 9022 11-03-1 have been performed.

The laser system complies with RoHS 2011/65/EU directive and CZ-Norm 5020.

3. Safety

Glucoloop's Ivy is a Class 3b laser. Direct high-peak-power, invisible light pulses can be hazardous to eyes. The laser power emitted by the Ivy may cause corneal damage. The skin of the human body can be irritated when exposed to the radiation. Therefore, care should be exercised during laser operation. Only persons familiar with the safety precautions and practices in this manual should operate the laser.

Direct exposure of the eye to the invisible laser beam must be avoided. Proper eye-wear must be worn at all times.

DANGER
AVOID EYE AND SKIN EXPOSURE TO LASER RADIATION
RADIATION OF THE IVY IS INVISIBLE.

CAUTION
Use of controls or adjustments or performance of procedures other

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than those specified herein may result in hazardous radiation exposure.

Personnel operating the laser product should be familiar with and follow the recommendations given below and in this manual:

1. Always wear laser safety eye-wear which is optically dense at the wavelength of operation.
2. Use safety interlocks to restrict access to laser areas and prevent unauthorized use of the lasers.
3. Never watch the laser light directly by eye.
4. Ensure that the laser light is always constrained in a dedicated and protected area so that accidental eye or skin exposure is not possible.
5. Do not use the IVY in a manner not specified by Glucoloop.
6. Do not open the laser.
7. Understand and follow all warning and safety labels.

The IVY is equipped with a safety interlock system (see Chapter 7). In case of an emergency, the laser can be fully switched off (no laser emission) by pulling the interlock line.

The optical parameters of the laser are (Laser Class 3b):

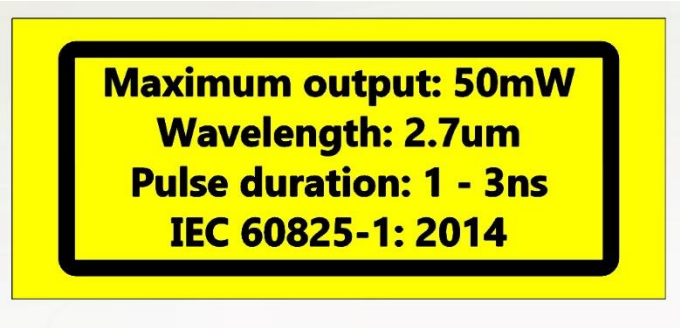
- Wavelength: 2720 nm +/- 100nm
- Beam divergence (full angle): <2 mrad
- Beam diameter: < 2.5 mm
- Pulse duration: < 3 ns
- Pulse repetition rate: ≤ 1.5 kHz
- Maximum average power: < 50 mW
- Maximum pulse energy: < 35 µJ

4. Laser warning labels and labels

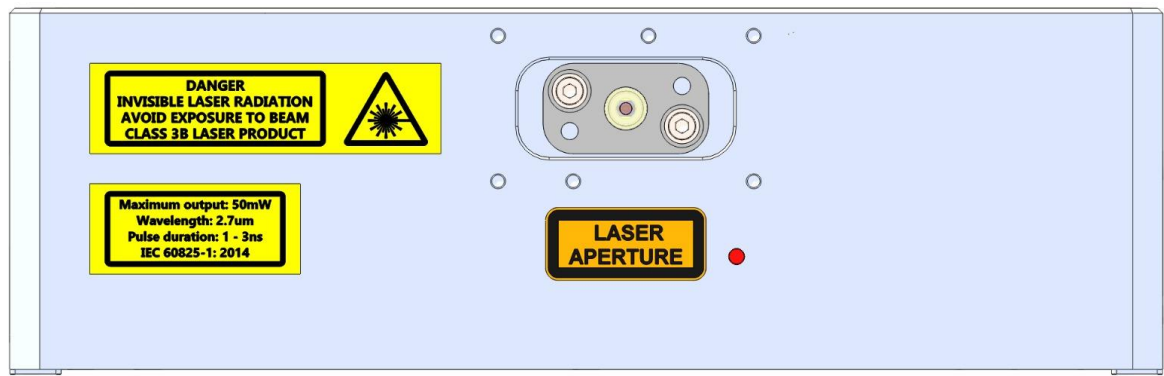
According to laser safety regulations the following **class 3b** laser product explanatory label with relevant laser parameters is used. For the detailed laser parameter please see the test report.



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Position of the laser warning labels on the laser housing:

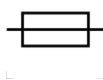
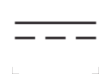


Nameplate on the laser:



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Symbol description table:

Symbol	Reference	Title
	Figure 14 of IEC 60825 – 1	Warning label - Hazard symbol Radiation of laser apparatus
	Figure 15 of IEC 60825 – 1	Laser aperture explanatory label
	IEC 60417 – 5032	Alternating current
	ISO 7000 – 0434A	Caution
	IEC 60417 – 5016	Fuse
	IEC 60417 – 5031	Direct current
	IEC 60417 – 5017	Earth, ground

5. Unpacking and Inspection

The Ivy is shipped in a package designed to provide maximum protection. If the shipping carton is damaged during shipping, immediately notify your shipping department, who may want to contact the carrier. Save the shipping box for future transportation or storage needs.

Remove and identify all of the following components:

- Ivy laser
- 24V power supply

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- Interlock connector
- Memory stick containing electronic copy of
 - Graphical User Interface executable file
 - Performance sheet of delivered unit
 - Certification of Conformity
 - User manual
- USB cable
- Dry air purging unit (optional)
- RS232 cable (optional)
- LAN cable (optional)
- Options

No other parts need to be examined or supplied by the manufacturer.

6. Operating conditions

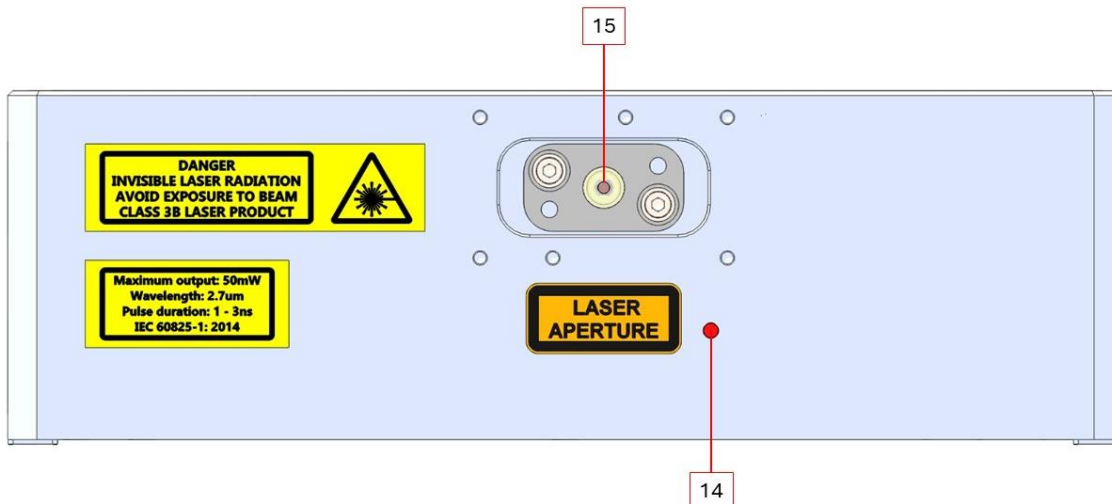
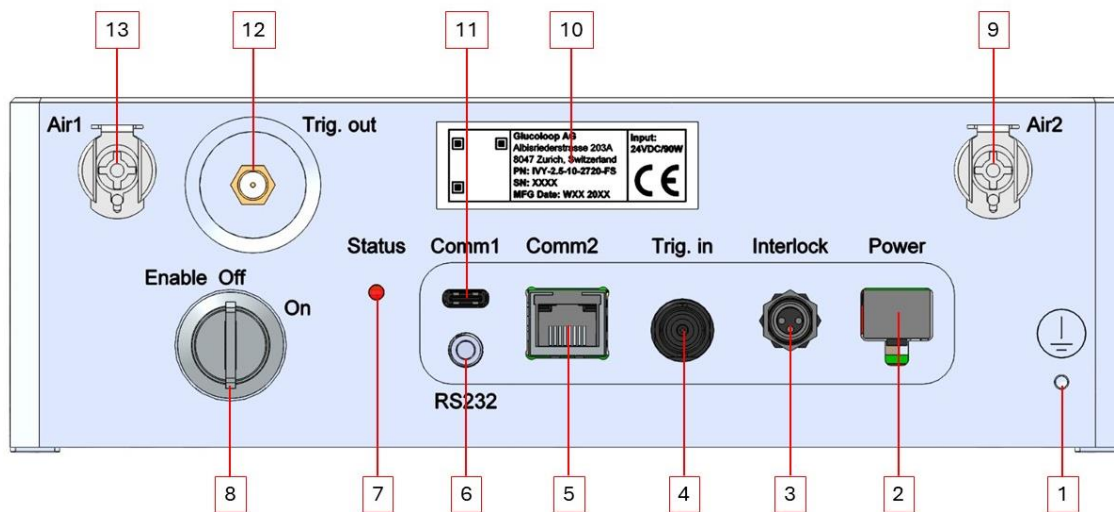
Electrical requirements:	24 VDC, 3.75A – see recommended power supply (Chapter 15)
)	
Power supply:	100 - 240V~ / 50-60Hz / 90W
Operating temperature:	15 - 35°C ambient
Storage temperature:	-20°C - 55°C
Humidity:	When purged with dry air: < 95% relative humidity
	Unpurged: < 40% relative humidity. A warning will be issued when humidity is higher.
Warm-up time:	Up to 10 minutes for laser emission, depending on environment
Air pressure operation:	700 hPa to 1060 hPa
Air pressure transport:	500 hPa to 1060 hPa
Pollution degree:	2
Installation category:	II

DO NOT STORE OR OPERATE THE IVY OUTSIDE THE SPECIFIED RANGE!

7. Connectors / Interfaces

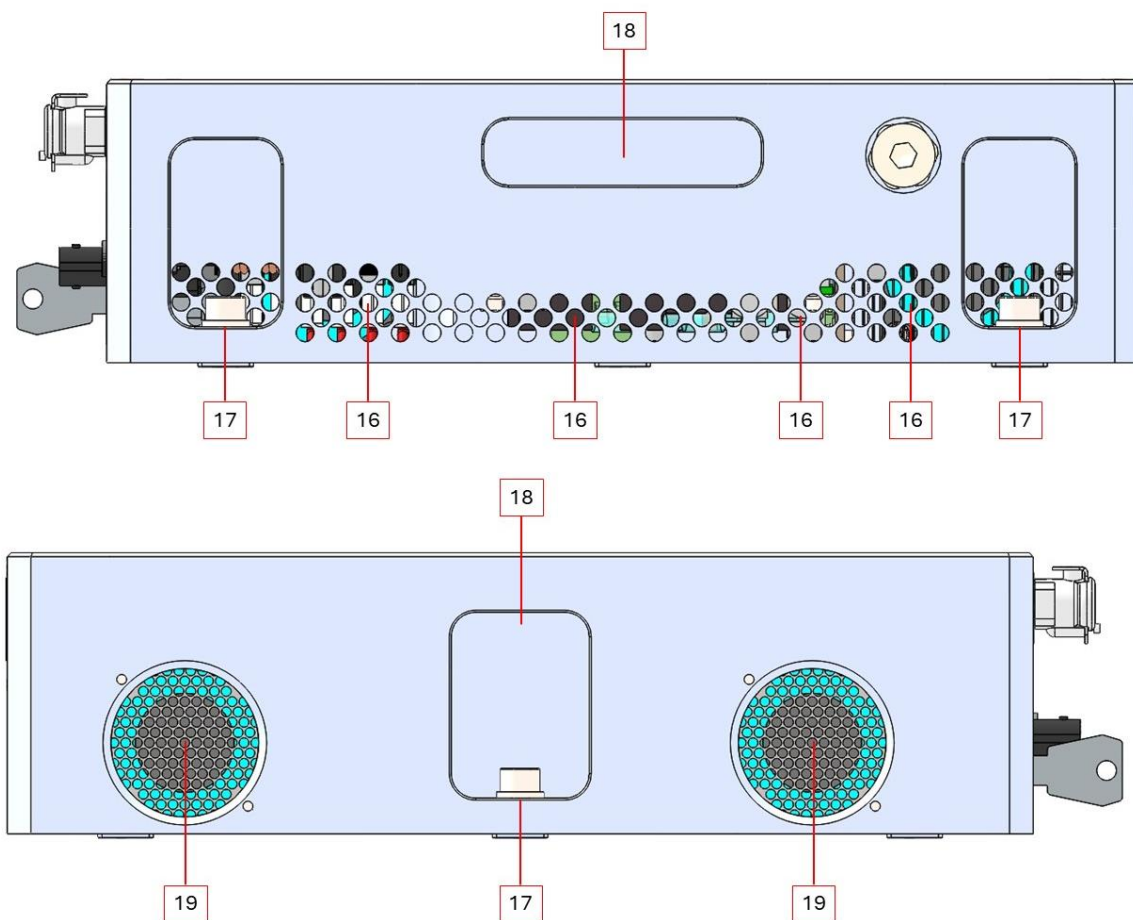
1. Grounding point
2. Power supply (Molex 39-01-2060 with contacts Molex 39-00-0077; Pin 1-3: 24V, Pin 4-6: Gnd)
3. Interlock (M8 3-pin female connector)
4. Trigger input (BNC)
5. LAN (RJ45)
6. RS232 (CUI Devices SJ1-43502PM, optional)

7. Status LED
8. Key switch
9. Dry air output (CPC PMCD1002)
10. Serial number label
11. Trigger output (SMA female, 50Ω)
12. USB (USB-C)
13. Dry air input (CPC PMCD1002)
14. Emission LED
15. Output window



8. Cooling

The Ivy is a forced air-cooled all-in-one laser package. An adequate air flow of up to 2.5 liters per second is required to ensure appropriate cooling and thus avoiding overheating of the device. Please make sure not to block the air inlet (16) and outlet (19) ports in order to ensure an adequate airflow through the laser. Please contact Glucoloop for further information should you design the laser into an instrument.



9. Mounting

The Ivy package contains two pockets along the long side of the package (18) which act as handles. Please grab the laser at these handles to transport it.

The package also contains three top-down through hole mounting holes (17). The mounting holes coincide with the designed area of support and ensure minimum stress in the laser package when

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mounted into a machine or onto an optical table. Please make sure that the mounting area is plane. If the laser is not screwed through the mounting holes but clamped, please use the same areas (17) to clamp the laser to its place. Do not apply mounting forces to other areas of the laser package. It could lead to deformation of the package and malfunctioning of the laser. Please make sure to well take care of the output beam area and to consider laser safety standards.

10. Parallel ports

The Ivy is equipped with a USB and an Ethernet port for communication and control of the laser. Glucoloop provides a simple graphical user interface (GUI) for the laser to be controlled (see Chapter 0), if used as a stand-alone device. Alternatively, and when integrating the laser into another device, the laser can be controlled directly via binary coded register-based communication. Please inquire with Glucoloop for details of the communication protocol.

11. Serial port

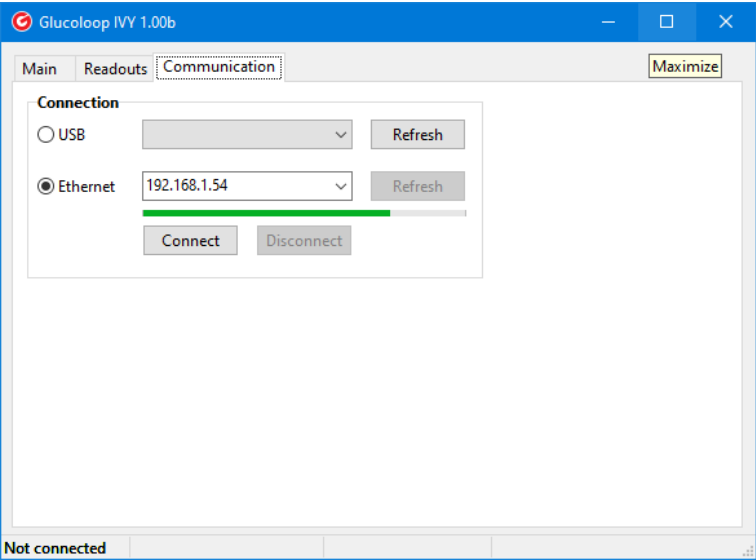
The Ivy is equipped with an RS232 serial communication port. Communication protocol is via binary coded register-based communication. Please inquire with Glucoloop for details of the communication protocol.

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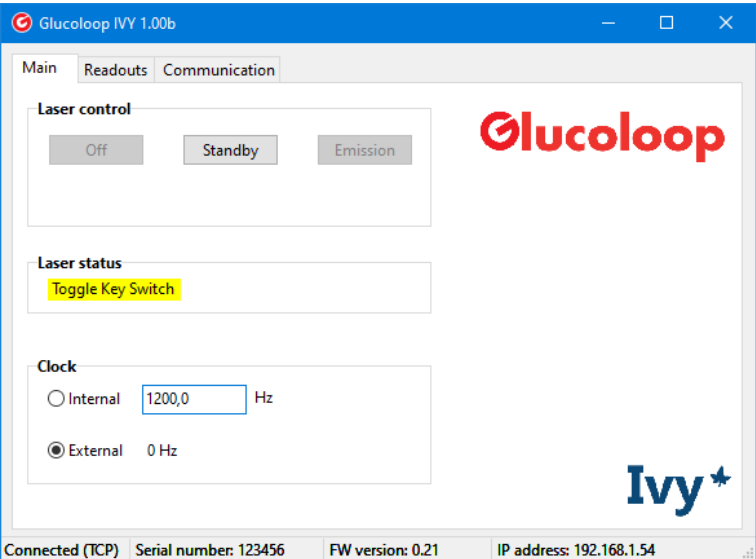
12. Graphical user interface (GUI)

The Ivy is provided with a simple GUI. The GUI contains 3 main tabs:

- **Communication** tab for setting up and controlling communication with the laser

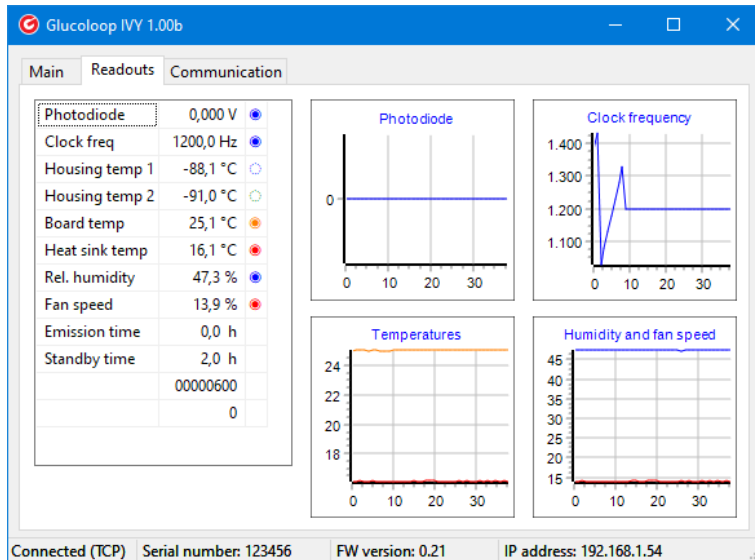


- **Main** tab for the daily use of the laser



The main tab also contains a “Laser status” area, where messages from the laser are displayed. This can be status messages or error messages.

- **Readout** tab for graphical representation of various readings from the laser



Please follow Section 19 for how to use and set-up the laser using the GUI.

13. Interlock

The laser is equipped with an interlock connector. Unless pin 1 and pin 3 of the connector are connected, laser emission is disabled. This interlock connector should be connected to the user's / integrator's safety circuit.

14. LEDs

The Ivy is equipped with a status LED (Chapter 7, item 7) and an emission LED (Chapter 7, item 14). The status LED, located next to the I/O panel of the laser, provides information about the status of the laser as follows:

- Flashing orange: Laser has just been powered up, and is searching for internal modules. This should take less than a few seconds.
- Slowly, gradually pulsing (breathing) green: Laser is in off state. No thermalization or emission is happening. Laser is ready for receiving commands.
- Fast flashing green (2 Hz): Laser is thermalizing. That can take up to 10 minutes.
- Solid green: Laser is thermalized and ready for emission or emitting (see emission LED and GUI status).
- Solid red: General error
- Red blinking once per second: Error with internal laser diode driver module
- Red blinking twice per second: Error with temperature control unit

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The emission LED is located close to the emission window of the laser. It is shining solid red, in case laser emission is enabled and class 3b laser radiation must be expected to exit from the housing.

15. Power supply

The Ivy is shipped with a sealed single 24V output 90W power supply, equipped with a standard IEC320-C14 inlet and adopting to the input voltage in the range of 90VAC to 264VAC. The power supply contains an overload protection (hiccup mode), an over voltage protection (re-power required) and an over temperature protection (re-power required).

In case the user provides its own power supply, the 24V supply cable from the users power supply to the laser must comply to IEC 60228, DIN VDE 0295, DIN VDE 0250, EN55011 and DIN VDE 0874 part 11. Fulfilling electrical safety (EN 61010-1:2010) and EMC (EN 61326-1:2013) requirements strongly depend on the used power supply. Glucoloop's certificate of conformity is only applicable if the delivered power supply is being used to power the laser.

16. Grounding the laser

In order to ensuring that the Ivy is complying with ESD and EMC requirements, the laser needs to be properly grounded. It is recommended to fix a 10 mm² copper wire or a 1 mm x 10 mm flat band cable with a 10 mm in diameter cable lug together with a M3 x 6 stainless steel screw on the earth point of the laser (Chapter 7, item 1).

17. Trigger output

The laser is equipped with a trigger output signal (Chapter 7, item 11). The connector type is a SMA type connector and the impedance is 50 Ohm. The trigger signal originates from an internal photodiode amplification circuit and thus is proportional to the pulse repetition rate of the laser.

18. Trigger input

Within the range of acceptable frequencies (see performance sheet of the laser), the Ivy can be triggered by an external clock. For that, a TTL signal of the proper frequency should be provided to the laser's trigger input port (Chapter 7, item 4). The connector type is a BNC type connector and the impedance is ≥ 50 Ohm. The external trigger input trig level: 0.5 V and the safe input voltage range is -6.0 to 6.0 V.

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19. Installation and getting started

1. Make sure that you and the working area are properly grounded. The Ivy laser contains components that are sensitive to electro-static discharge. Non-appropriate grounding might lead to permanent damage of the device.
2. Place the laser where you want to operate it. Take care not to block the ventilation holes. The required air flow through the laser is up to 2.5 liters per second. Fix the laser with the three through-hole screws or clamp it at the screw positions in its place. Make sure the output area is well secured (Chapter 8, item 17) and the laser is properly grounded. Make sure, that the output beam will not be reflected back into the laser – this can lead to permanent damage.
3. Make sure the output window (Chapter 7, item 15) is clean.
4. We suggest that the laser is purged with clean (filtered) dry air via the air inlet (Chapter 7, item 13) and air outlet (Chapter 7, item 9). As the laser is operating close to the absorption peak of water, stability of the laser is significantly increased when purged with dry clean air. Please make sure not to insert any dust, oil or other contaminants through the self-closing purging fittings. Glucoloop is offering a well-fitting purger unit as an option, however any purger unit providing filtered 0.25-5 liters of clean dry air per minute can be used.
The Ivy contains an internal humidity sensor. If the humidity exceeds a certain level, there will be a warning message provided from the laser and displayed in the GUI. If the humidity exceeds a further threshold, which is relevant for stable operation, it will shut down emission and temperature control, and provide an according message.
5. Connect a the laser via Ethernet (Chapter 7, item 5item 5) or USB (Chapter 7, item 12) to a computer.
6. Make sure the interlock circuit is closed (Chapter 7, item 3). The delivery contains a fitting connector with appropriate pins connected. Please use a connector of the same type (M8 3-pin female) and connect it to your interlock system. When pin 1 and pin 3 are shortened, laser emission can be enabled. When they are open, laser emission is disabled.
7. Connect the 24V power supply to the laser head (Chapter 7, item 2) and to the power line (100-240 V; 50-60 Hz). The laser will check all internal modules for proper communication and status, in this period the status LED (Chapter 7, item 7) is blinking in orange. Once all internal modules are cleared the laser is ready to receive commands and the status LED is breathing slowly in green.
8. Start the executable GUI.exe by double clicking on the according icon. It is included in the delivery on to the memory stick.
9. Navigate to the “Communication” tab of the GUI to establish connection to the laser.
If you have connected the laser via Ethernet, please press the “Ethernet” selection button and push the “Refresh” button. The GUI will perform an IP scan on the network, the progress of which is shown in the green progress bar. Once the scan is completed the compatible lasers on your network will be shown in the Ethernet drop down list and you can select the appropriate laser. By pushing the “Connect” button, the GUI will establish a communication connection to the laser’s main controller.
If you chose to connect the Ivy via USB to the computer, select the USB radio button and then the “Refresh” button. You can now chose the appropriate COM port in the USB drop-down list. Then

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push the “Connect” button and the GUI will establish a communication connection to the laser’s main controller.

A successful connection will be shown in the bottom left bar of the GUI window.

10. Go to the “Main” tab of the GUI. In the “Clock” area, select if you want to use the internal clock of the laser, or if the laser should be triggered externally. Make sure that the frequency is within the range of allowable frequencies for the corresponding laser unit. If the frequency is outside the allowable frequency range (see performance sheet), laser emission will not be enabled.
11. Toggle the key switch (Chapter 7, item 8)
12. Set laser to standby, by pressing the “Standby” button. In this mode the laser will thermalize, i.e. all components are brought to the defined temperature. During thermalization, the status LED is flashing green at a frequency of about 2 Hz and the GUI is displaying a spinning wheel waiting symbol with a “Stabilizing” label. This thermalization process can take up to 10 minutes, depending on the ambient conditions. Once thermalization is complete, the status LED will shine firmly green and the laser is ready for emission. The laser is already rated laser class 3b in this mode!
13. In order to enable laser emission, press the “Emission” button. Laser emission will occur. The emission LED (Chapter 7, item 14) will shine up firmly in red and the GUI will show emission in the bottom bar and in the “Laser status” area, together with a laser beam symbol.
14. The pulse repetition rate can be controlled while the laser is emitting.
15. In order to stop emission of the laser, press the “Standby” button or the “Off” button, depending on what you intend to do later. Once the laser is in “Off” state, the power supply can be detached, if required.

20. Operation and daily use

The Ivy is designed for operation environments from 15 – 35 °C. If relative humidity exceeds 40% the laser must be purged with clean dry air. For increased stability we recommend to purge the laser with clean dry air in any case.

21. Storing the laser

If not in use, the Ivy should be stored in a clean and dry environment at room temperature.

22. Error states

Please inquire with Glucoloop for details of list of the codes of the error states.

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23. Maintenance and Service

The IVY is designed to be maintenance free. No further calibrations, adjustments or “tweaks” are necessary for the IVY to meet its performance specifications. No service of optical parts can be carried out by service personnel outside Glucoloop’s facilities.

In case of failure of the product, please contact Glucoloop directly:

admin@glucoloop.com, Tel. +41 31 503 58 98

24. Warranty

Glucoloop warrants the products to the original purchaser (the buyer) only. Glucoloop warrants that the laser system of this delivery (including all options), is free from defects in materials and workmanship.

Glucoloop warrants the laser system to be free from defects in materials and workman-ship for a period of 1 year after shipment of the laser.

Responsibilities of the Buyer

The Buyer is responsible for providing the appropriate utilities and an operating environment as outlined in the product literature. Damage to the laser system caused by failure of Buyer’s utilities or failure to maintain an appropriate operating environment, is solely the responsibility of the Buyer and is specifically excluded from any warranty, warranty extension, or service agreement. The Buyer is responsible for prompt notification to Glucoloop of any claims made under warranty. In no event will Glucoloop be responsible for warranty claims made later than seven (7) days after expiration of warranty.

Limitations of Warranty

The foregoing warranty shall not apply to defects resulting from:

- Components and accessories manufactured by companies, other than Glucoloop, which have separate warranties
- Improper or inadequate handling by the Buyer
- Buyer-supplied interfacing
- Operation outside the environmental specifications of the product
- Unauthorized modification or misuse
- Improper site preparation and maintenance
- Attempt of opening the housing of the laser

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Glucoloop assumes no responsibility for customer-supplied material. The obligations of Glucoloop are limited to repairing or replacing, without charge, equipment which proves to be defective during the warranty period. Replacement sub-assemblies may contain reconditioned parts. Repaired or replaced parts are warranted for the duration for the original warranty period only.

Warranty applies only to the original purchaser, unless otherwise specified in the sales contract. Warranty is transferable to another customer only by special agreement with Glucoloop.

In case of warranty event, contact Glucoloop in order to get an RMA number assigned. Use the original packing material for returning the laser to Glucoloop and label it properly as fragile. Follow the instructions given by Glucoloop. In case of not following these instructions any damage that could have occurred during shipment will be excluded from warranty.

25. Drawings



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